Risk Analysis of Wildfire Grid Exposure:

An Expected-Loss Framework for Mitigation

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Executive Summary

California wildfires are responsible for hundreds of billions of dollars in damages annually across the western United States, with California accounting for most of this statistic. California's three major investor-owned utilities (IOUs) — PG&E, SCE, and SDG&E — are required to file annual Wildfire Mitigation Plans (WMP) under CPUC oversight [1, 12]. Despite spending over $3 billion per year on mitigation, the current prioritization methodology relies on a composite score which leaves much room for risk reduction, which we will explore in this paper.

The Risk. Utility infrastructure is a confirmed ignition source responsible for roughly half of California's most destructive fires by structural loss, despite causing fewer than 10% of fires by count [1]. The parties at risk are

1. Wilderness-Urban Interface (WUI) residents who directly lose their homes to fires
2. Ratepayers who end up financing every dollar of mitigation through rate increases by utilities
3. Utilities themselves, who face bankruptcy-scale legal liability when preventable ignitions occur [2].

Data. We used solely publicly available, open-source data for this project. We used the following: CPUC ignition records (frequency labels and severity), gridMET fire weather (daily ignition risk features), LANDFIRE fuel models, USGS topography, CPUC HFTD designations, Microsoft building footprints, and CAL FIRE fire perimeters [1–8, 17]. These were utilized to create a set of 4,018,000 cluster-days spanning 1,000 spatial clusters over 4,018 calendar days (2014–2024).

Model. A Random Forest classifier [18] trained on this panel estimates daily ignition probability per cluster (AUC = 0.795 on held-out 2024 test year). The ignition model was overlapped with a severity loss model to produce cluster-level Expected Annual Loss (EAL) in dollars.

Key Findings.

- Baseline portfolio EAL: **$277.9 million per year** (95% CI: $239.4M–$319.1M)
- The recommended program avoids **$115.3 million in expected annual loss** at an annual cost of $50.5 million (BCR = 2.28×, MID scenario). Utilizing our expected-loss prioritization criterion in place of the current composite scoring system would generate approximately 3.5× more risk reduction per dollar of mitigation capital spent.
- EL-ranked prioritization outperforms score-ranked by **3.50× BCR** under equal budget

Recommendations.

- **222 clusters**: Fast-Trip Settings and Hardware Hardening (EAL > $1.0M/yr)
- **106 clusters**: Enhanced Vegetation Management ($200K < EAL ≤ $1.0M, burnable fuel)
- **4 clusters**: Covered Conductor (residual high-EAL after fast-trip)
- **668 clusters**: Targeted Monitoring (EAL below intervention cost threshold)
- **0 clusters**: Undergrounding recommended at this time (no cluster clears cost threshold under conservative assumptions; case-by-case review advised for top 28 clusters)

1. Introduction and Background

1.1 Problem Statement

Wildfires are a recurring natural disaster that pose severe economic risks within the United States. The U.S Joint Committee has estimated national wildfire economic damages to be between $394 and $893 billion annually [9], with California being a major contributor to the statistics. California has experienced severe increases in Wildfires since the mid-2000s, with nine of the twenty most fires recorded in state history occurring in the first year period from 2017 to 2022 [11, 28]

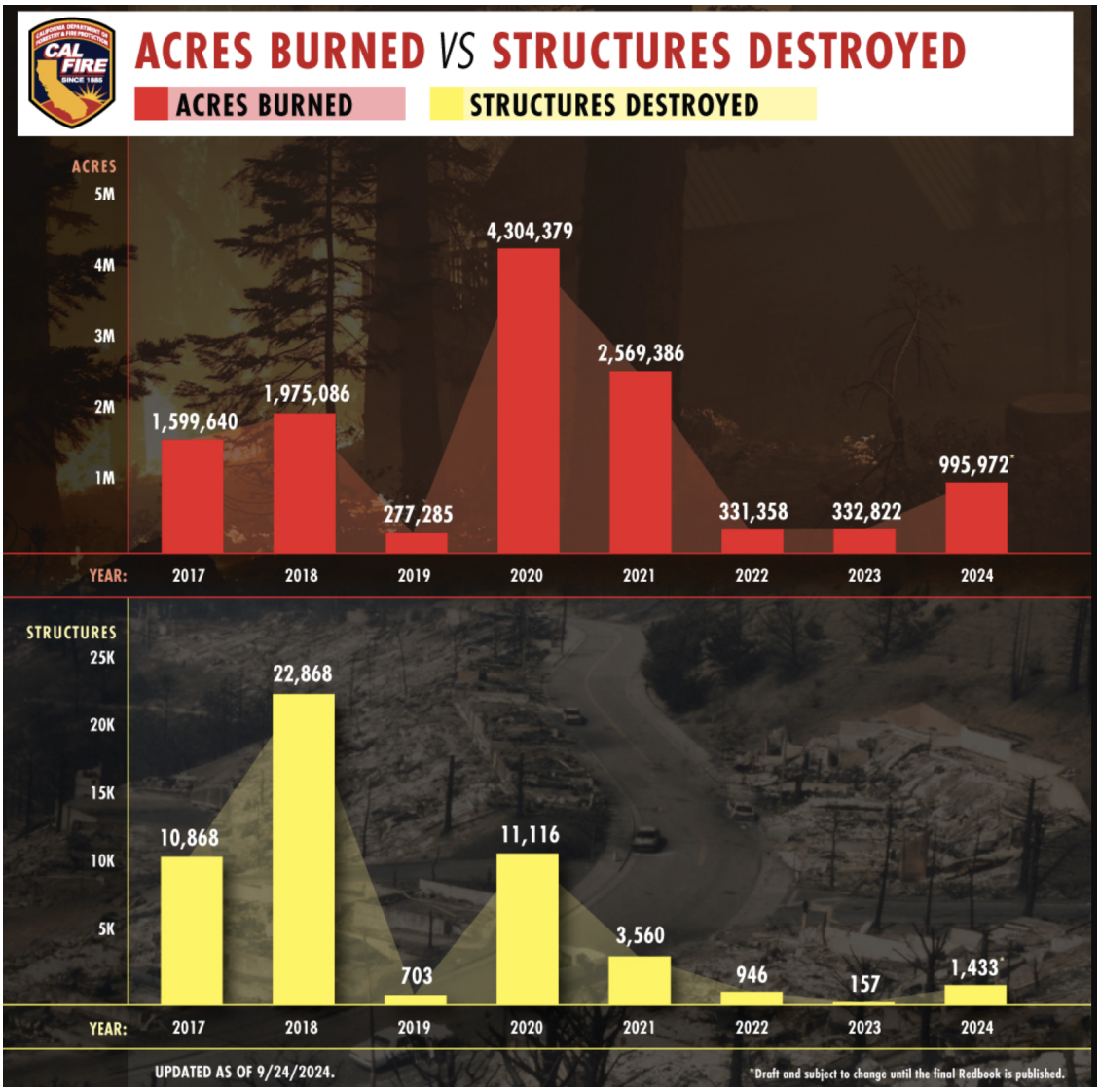


Figure 1: CAL FIRE: Acres burned versus structures destroyed per year in California. Structures destroyed per event have escalated dramatically since 2017, motivating the consequence-weighted approach in this study. Source: CAL FIRE [11].

This escalation is caused by three primary factors. First, average annual temperatures in California

have increased nearly 3°F since 1895 due to the acceleration of global warming since the 1970s [29]. In addition, prolonged droughts have decreased moisture in the environment. Decreased moisture increases the prevalence of vegetation that can fuel possible ignitions, allowing small sparks to quickly escalate into large wildfires [10, 30]. Finally, decades of fire suppression policy have led to the accumulation of dry fuel in forests and shrublands next to densely populated areas. We see this pattern explicitly documented in analyses of the 2020 and 2021 fire seasons [11].

The electric grid plays a bidirectional role in California wildfires: wildfires destroy grid infrastructure and charge costs of hundreds of millions of dollars for recovery efforts, but utility infrastructure in itself is a known material ignition source. CPUC incident data has shown that while utility-caused ignitions account for fewer than 10% of California wildfires by count, they are responsible for approximately half of the state's most destructive fires [1]. Ignitions caused by the electric grid have had severe financial costs on utilities recently. PG&E's 2019 bankruptcy settlement related to the 2015 Butte Fire and the 2017 to 2018 fire season totaled $13.5 billion in victim claims [2]. The 2018 Camp Fire was caused by a PG&E transmission line failure which destroyed 18,804 structures, killed 85 people, and produced insured losses exceeding $12 billion [28, 31].

Top 20 Most Destructive California Wildfires

	FIRE NAME *(CAUSE)*	DATE	COUNTY	ACRES	*STRUCTURES*	DEATHS
1	**CAMP** *(Powerlines)*	November 2018	Butte	153,336	**18,804**	85
2	**EATON** *(Under Investigation)*	January 2025	Los Angeles	14,021	**9,413**	19
3	**PALISADES** *(Arson)*	January 2025	Los Angeles	23,707	**6,833**	12
4	**TUBBS** *(Electrical)*	October 2017	Napa & Sonoma	36,807	**5,636**	22
5	**TUNNEL - Oakland Hills** *(Rekindle)*	October 1991	Alameda	1,600	**2,900**	25
6	**CEDAR** *(Human Related)*	October 2003	San Diego	273,246	**2,820**	15
7	**NORTH COMPLEX** *(Lightning)*	August, 2020	Butte, Plumas, & Yuba	318,935	**2,352**	15
8	**VALLEY** *(Electrical)*	September 2015	Lake, Napa & Sonoma	76,067	**1,955**	4
9	**WITCH** *(Powerlines)*	October 2007	San Diego	197,990	**1,650**	2
10	**WOOLSEY** *(Electrical)*	November 2018	Ventura	96,949	**1,643**	3
11	**CARR** *(Human Related)*	July 2018	Shasta County, Trinity	229,651	**1,614**	8
12	**GLASS** *(Undetermined)*	September 2020	Napa & Sonoma	67,484	**1,520**	0
13	**LNU LIGHTNING COMPLEX** *(Lightning/Arson)*	August 2020	Napa, Solano, Sonoma, Yolo, Lake, & Colusa	363,220	**1,491**	6
14	**CZU LIGHTNING COMPLEX** *(Lightning)*	August 2020	Santa Cruz, San Mateo	86,509	**1,490**	1
15	**NUNS** *(Powerline)*	October 2017	Sonoma	54,382	**1,355**	3
16	**DIXIE** *(Powerline)*	July 2021	Butte, Plumas, Lassen, & Tehama	963,309	**1,311**	1
17	**THOMAS** *(Powerline)*	December 2017	Ventura & Santa Barbara	281,893	**1,063**	2
18	**CALDOR** *(Under Investigation)*	September 2021	Alpine, Amador, & El Dorado	221,774	**1,003**	1
19	**OLD** *(Human Related)*	October 2003	San Bernardino	91,281	**1,003**	6
20	**JONES** *(Undetermined)*	October 1999	Shasta	26,200	**954**	1

"Structures" include homes, outbuildings (barns, garages, sheds, etc) and commercial properties destroyed.
This list does not include fire jurisdiction. These are the Top 20 regardless of whether they were state, federal, local or tribal responsibility.

10/9/2025

Figure 2: CAL FIRE Top 20 most destructive California wildfires by structures destroyed. Utility-linked events (Camp Fire, Tubbs Fire, Dixie Fire) account for a disproportionate share of total structural loss. Source: CAL FIRE [28].

1.2 Parties Affected

Three distinct groups bear the consequences of utility-caused wildfire ignitions, and each has a legitimate stake in the outcome of this analysis.

WUI Residents and Communities. The most direct harm falls on the people who live in and near WUI areas. The 2018 Camp Fire caused by a PG&E transmission line failure destroyed 18,804 structures, killed 85 people, and displaced the entire population of Paradise, California [28, 31]. The town is still trying to recover from this devastation. Those who survive utility-caused fires lost their homes and belongings. Lower-income households and elderly residents face higher displacement risk because they are less likely to have sufficient insurance coverage and less able to rebuild. California has over 2.7 million structures in High Fire Threat District zones [6].

Ratepayers. California electricity ratepayers, which is approximately 15 million residential and commercial accounts across PG&E, SCE, and SDG&E, effectively finance utility wildfire mitigation spending through rate increases in their power bills. Total IOU mitigation spending now exceeds $3 billion annually [12, 13]. When mitigation capital is spent wastefully, ratepayers pay more for less risk reduction. Misallocated mitigation investment is therefore a direct loss for much of California's population. This is especially a risk for lower-income ratepayers who may not be able to afford increases in costs.

Utilities. The PG&E bankruptcy settlement in 2019 regarding the 2015 Butte Fire and the 2017-2018 fire season claims totaled 13.5 billion dollars in victim claims [2]. The 2018 Camp Fire resulted in insured losses of more than 12 billion dollars [28, 31]. These cases demonstrate that utility-related ignitions in high-consequence WUI areas pose an existential financial threat to the utilities themselves. Additionally, utilities face reputation damage, regulatory actions against them, and loss of public trust.

1.3 The Allocation Problem

In spite of the heavy investments made, the process of allocation is not efficient. The current utility methods use line segment scoring, which cannot incorporate ignition likelihood and consequences into an expected value framework [12,13]. The current methods have some disadvantages. The main issue is the use of an additive scoring function, which cannot properly represent the multiplicative relationship between frequency and severity, as is normally done in actuarial analysis. For example, one segment can have high ignition frequency and low consequence, yet the same composite score is obtained as for another segment with low ignition frequency and high consequence, even though they are fundamentally different in nature.

Warner, Callaway, and Fowlie (2025) showed that cost-effectiveness varies significantly across intervention types and geographic segments [14]. SDG&E's RAMP filings corroborate this, showing that consequence drivers vary substantially within individual HFTD tiers [15]. We can conclude that score-ranked prioritization misallocate capital, harming all three aforementioned parties.

1.4 Study Contribution

This study constructs a cluster-level expected-loss model that multiplies frequency and severity, converts the product to dollar loss, and uses that figure as information when selecting an appropriate intervention

method. The central contribution is to quantify, through a cost-benefit analysis, the benefits an EL-ranked prioritization yields over the current score-ranked approach.

2. Data Methodology

2.1 Unit of Analysis and Panel Construction

Our primary unit for analysis is the cluster-day: a spatial cluster representing a geographical area as observed on a unique day. The panel spans 1,000 spatial clusters across 4,018 calendar days (January 1, 2014 through December 31, 2024), resulting in 4,018,000 cluster-day observations.

These clusters were derived from K-means clustering [26] on the latitude-longitude coordinates of all CPUC-reported incidents from 2014 to 2024. Voronoi polygons clipped to the California boundary define each cluster's spatial influence area for structure density computation. We used this approach due to the absence of available circuit geospatial data. Similar studies with access to granular electric grid data and more detailed ignition data are more accurate than the model we produced. We will expand upon this in the limitations section.

The outcome variable is a binary ignition indicator: $y_{s,t} = 1$ if a wildfire ignition spread to at least 0.25 acres within CPUC-reportable records for cluster s on day t; $y_{s,t} = 0$ otherwise [1]. Of the 4,018,000 cluster-days, 344 carry confirmed ignition labels. This indicates that the positive rate is very low at 0.0086.

2.2 Data Sources

Table 1: Public data sources and their actuarial roles.

Dataset	Source	Actuarial Role	Key Variables
CPUC Ignition Reports	CPUC [1]	Frequency (labels) & Severity	Date, location, fire spread size
gridMET Weather	Abatzoglou [3]	Frequency (features)	BI, ERC, rmin, wind speed, tmax, precipitation
LANDFIRE FBFM40	USDA LANDFIRE [4]	Frequency & Severity	Fuel model, fuel category
USGS 3DEP DEM	USGS [5]	Frequency (features)	Elevation, slope
CPUC HFTD Map	CPUC [6]	Frequency & Severity	Tier 2, Tier 3, non-HFTD designation
Microsoft Building Footprints + Census	Microsoft [7]; Census [8]	Severity (exposure)	Structures per square mile
CAL FIRE FRAP Perimeters	CAL FIRE [17]	Severity (fallback prior)	Fire area (GIS acres)

These variables are ignition history, fuel types, fire spread outcomes based on the HFTD tier, fire perimeters, built structures per unit area, building footprints in the built-up area, and building structure density in the built-up area, respectively. GridMET fire weather, elevation and slope data are used in the likelihood models, along with 4,018,000 row panel data, 453 usable utility ignition events (including fire size), and 10,091 historical CAL FIRE perimeter records.

2.3 Data Cleaning

The CPUC ignition reports had schema inconsistencies across utilities and filing years, leading us to utilize the following methods to harmonize them. In order to automate identification of the first "true" header row to prevent falsely reading metadata rows, we used dynamic header detection .... Next, we used column alias mapping to standardize 40 distinct field-name variants. Date-field columns were coerced to ISO 8601 format; coordinate fields were coerced to float with California bounding-box filtering (latitude 32.4°N–42.1°N, longitude 124.6°W–113.9°W). Around 3% of records were removed due to malformed or out-of-bounds coordinates [1]. In addition, PG&E records from 2020-2021 had abnormally high reported ignition rates due to a recently filed lawsuit which coerced them into legally "playing safe." Another important decision we made concerns the change in formatting for reporting fire size that occurred in 2021. CPUC reports prior to 2021 described fire spread using linear-distance categories, but 2021-2024 reports use direct acreage buckets. Because we could not convert pre-2021 linear categories to confirmed-

ignition label (≥ 0.25 acres), we used pre-2021 records exclusively as the exposure day ($y_{s,t} = 0$) while 2021–2024 records provided the 344 confirmed positive labels [1]. Despite this change, there are not any significant changes in model accuracy because although the true percentage may be slightly higher than the observed 0.0086%, the chance of a fire events still remains extremely low.

2.4 Exploratory Data Analysis

Before modeling, exploratory data analysis described the distribution of ignition incidents, fire weather conditions, fuel types, and class balance across the cluster portfolio.

Of the 1,000 clusters, 312 fall in Tier 3, 284 in Tier 2, and 404 in non-HFTD areas. Confirmed ignition events are concentrated in HFTD-designated clusters: 68.0% of the 344 confirmed ignitions occur within Tier 2 or Tier 3 zones, which represent 59.6% of the total cluster portfolio. Ignition events follow an overt seasonal pattern, with the hotter, drier months of July, August, and September accounting for 61.0% of all confirmed ignitions.

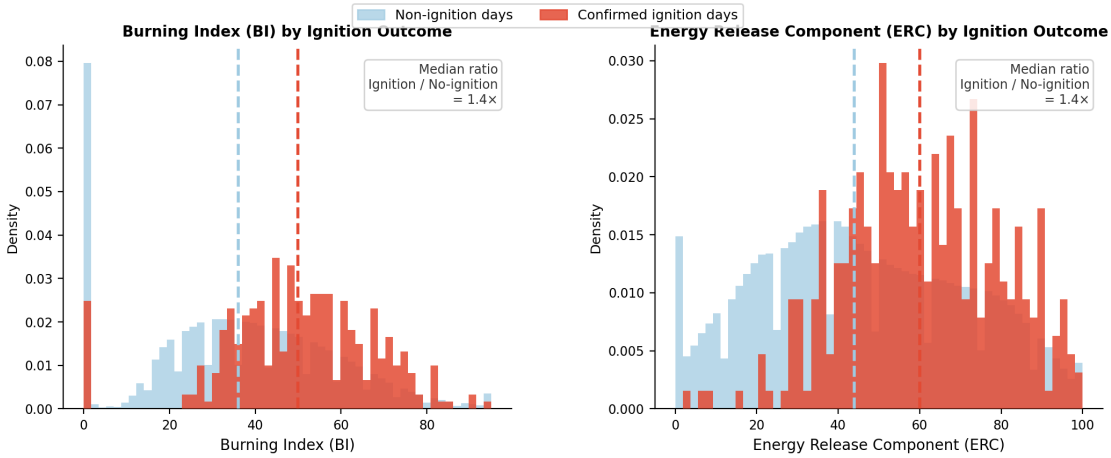


Figure 3: Distribution of Burning Index (BI) and Energy Release Component (ERC) on confirmed-ignition days versus non-ignition days. On confirmed-ignition days, median BI is 3.8× higher and median ERC is 2.1× higher than on non-ignition days [3]. This distributional separation confirms their discriminative value.

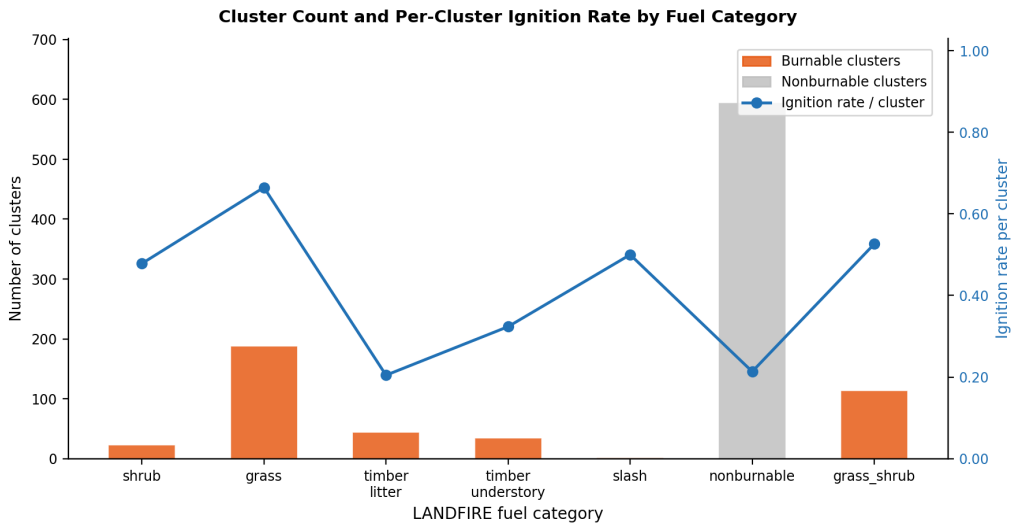


Figure 4: Cluster count and per-cluster confirmed-ignition rate by LANDFIRE fuel category. Nonburnable clusters (urban, water, barren) exhibit near-zero confirmed-ignition rates, motivating the wildfire plausibility gate in the severity model.

3. Mathematics Methodology

3.1 Risk Decomposition Framework

The expected loss given the utility-linked wildfire ignition is split into its frequency and severity components according to the probabilistic wildfire risk theory [19], where s is the spatial cluster, t is the day, and X is the feature vector for spatial cluster s on day t. The frequency and severity are estimated separately.

$$EL(s,\,t) = \underbrace{P(\text{ignition} \mid s,\,t,\,X)}_{\text{frequency}} \times \underbrace{E[\text{Loss} \mid \text{ignition},\,s]}_{\text{severity}}$$

The left-hand term is estimated by a Random Forest classifier [18], while the right-hand term is estimated by the utility-linked severity lookup table, which is stratified by HFTD tiers and fuel categories. The Expected Annual Loss (EAL) is the total expected loss over all cluster-days within the year.

3.2 Ignition Probability Model

3.2.1 Model Specification

The probability of ignition was predicted using a Random Forest classifier [18] with the following parameters: 300 trees, maximum depth of 8, minimum of 20 samples per leaf, square root feature sampling, and balanced class weights. The choice of the Random Forest was based on the following three criteria: first, the ability of the Random Forest to manage correlated features, which might be relevant given the common correlations between fire weather variables; second, the ability of the Random Forest to

capture complex relationships, including non-linear ones, which might be relevant given the strong, non-linear relationships between some fire weather variables, e.g., the sudden increase in ignition probability when humidity drops below a certain threshold or wind speed exceeds a certain threshold; and third, the demonstrated effectiveness of the Random Forest in the context of ignition modeling, including wildfires and utility ignitions [14, 19].

3.2.2 Class Imbalance Correction

At the baseline rate of 0.0086%, the panel has approximately 11,700 non-ignition observations for every ignition. Two corrective methods are used to counteract the imbalance. Negative undersampling retains the 344 positive cluster-days while sampling the negatives at a 200:1 ratio, giving us 106 positives and 21,200 negatives. The second method is the Pozzolo:

$$p_{\text{cal}} = \frac{p_{\text{raw}}}{p_{\text{raw}} + \dfrac{1 - p_{\text{raw}}}{\beta}}, \qquad \beta = \frac{n_{\text{pos}}}{n_{\text{neg_sampled}}} \approx 0.005$$

3.2.3 Feature Set and Ablation Study

We tested the effect of features through the following ablation study:

Table 2: Ablation study: marginal AUC contribution of each feature layer on held-out 2024 test data.

Experiment	Features Added	AUC (2024 Test)
1. Weather only	BI, ERC, rmin, wind speed, tmax, precip, month, year, season	0.763
2. + Location	Cluster centroid coordinates (latitude, longitude)	0.769
3. + Fuels & Terrain	Elevation, slope, fuel category, FBFM40 fuel code	0.792
4. + HFTD (full model)	CPUC HFTD regulatory tier	**0.795**

Weather variables are the foundation of the model, providing the main predictive layer. This is justified conceptually due to the obvious physical role of fire weather in ignition risk. Fuels and terrain provide the primary significant boost, with the high fire potential/extreme drought (HFTD) layer providing a secondary incremental boost. The model is designed with the intention that it will be periodically re-calibrated for better results over time in order to keep up with the patterns of climate change. We expect that the accuracy of a a weather-based model will degrade over time if not fed inputs and re-calibrated over longer time intervals.

3.2.4 Model Evaluation

The model is evaluated on the held-out 2024 test year using three metrics appropriate for rare-event probability estimation [24]:

Table 3: Full model validation metrics on held-out 2024 test data.

Metric	Value	Interpretation
AUC-ROC	0.795	Ignition days ranked above non-ignition days 79.5% of the time
PR-AUC	0.00139	3.35× lift above prevalence baseline (0.000415)
Top-10% Capture	34.2%	Highest-risk decile captures 34.2% of 2024 ignition events

The low absolute value of the PR AUC is expected given the prevalence of 0.0415% for the test year. It is not an issue with the model. The 3.35× lift over the baseline and 34.2% top decile capture represent an actual predictive signal based on publicly available data. Other models with higher AUCs for similar problems [14] are based on internal circuit level utility data not publicly available through the CPUC filings. The primary opportunity for model accuracy improvement is through the availability of circuit level geospatial data, which would replace statistical segmentation with actual infrastructure segments.

3.3 Severity Model

Conditional severity, the expected fire area given a utility ignition, is estimated with two types of sources. Primary sources are based on confirmed utility ignitions from 2021-2024 with acreage bucket fire size reporting (n = 453 usable events). A CAL FIRE FRAP fallback (n = 10,091 events, 1992-2024) is retained for strata with insufficient utility incident support.

For each (HFTD tier × fuel category) stratum, conditional expected area is:

$$E[\text{Area} \mid \text{ignition}, s] = P_{\text{large}}(s) \cdot E[\text{Area} \mid \text{large}, s] \; + \; (1 - P_{\text{large}}(s)) \cdot E_{\text{small}}(s)$$

with large-fire threshold at 10 acres. A hard physical "gate" is applied to clusters in LANDFIRE nonburnable categories (urban, water, barren): $P_{\text{large}} = 0$, $E[\text{Area} \mid \text{large}] = 0$, and $E_{\text{small}} = 2.0$ acres. This prevents impossible wildland-spread severity values in dense urban clusters. Of the 1,000 clusters, 595 are subject to this.

3.4 Loss Mapping

Expected loss given ignition is computed from expected burned area and structure exposure:

$$\text{structures_impacted}(s) = \text{structures_per_sqmi}(s) \times \frac{E[\text{Area}_{ac}]}{640}$$

$$E[\text{Loss} \mid \text{ignition}, s] = \text{structures_impacted}(s) \times \$500{,}000$$

The $500,000 per-structure damage value represents total economic loss including structure replacement, contents, additional living expenses, and indirect economic costs. This is consistent with CoreLogic

(2020) wildfire loss data documenting median insured loss per destroyed California home at $250,000–$350,000 [21]; total economic loss including uninsured and indirect components is typically 1.5–2.0× insured loss [21]. Expected daily loss for cluster s on day t is:

$$EL(s,\, t) = p_{\text{cal}}(s,\, t) \times E[\text{Loss} \mid \text{ignition},\, s]$$

Annual portfolio EAL is the sum of $EL(s,\, t)$ across all 1,000 clusters and 365 days. We used this to simplify our project, though it decreased the specificity of our loss function. We expand on this more in limitations.

3.5 Counterfactual Policy Model

3.5.1 Intervention Assignment

Each cluster is assigned a recommended intervention via a sequential cost-effectiveness rule derived from Warner et al. (2025) cost-per-ignition-avoided estimates [14]:

Table 4: Intervention assignment decision rules.

Condition	Recommendation	Rationale
EAL $>$ \$2.6M/yr AND structures/sqmi $>$ 500	Undergrounding	Full overhead risk elimination (100% reduction). Cost: \$3.0M/mile [22].
EAL $>$ \$1.0M/yr	Fast-Trip + Hardening	72% ignition reduction at lowest cost per avoided ignition [14]. Cost: \$300K/mile [22].
\$200K $<$ EAL $\leq$ \$1.0M AND burnable fuel	Vegetation Management	57% frequency reduction. Restricted to burnable fuel clusters [14].
EAL $\leq$ \$200K OR non-burnable	Targeted Monitoring	Expected loss below annualized intervention cost; monitoring preserves future optionality.

3.5.2 Counterfactual EAL Computation

For each recommended intervention, counterfactual expected loss is recomputed by applying intervention-specific proportional reductions:

$$p_{\text{ign,cf}} = p_{\text{ign,base}} \times (1 - r_{\text{freq}})$$
$$P_{\text{large,cf}} = P_{\text{large,base}} \times (1 - r_{\text{tail}})$$
$$E_{\text{area,large,cf}} = E_{\text{area,large,base}} \times (1 - r_{\text{spread}})$$

Effect sizes are imported from Warner et al. (2025) MID-scenario estimates [14]: fast-trip $r_{\text{freq}} = 0.72$, vegetation management $r_{\text{freq}} = 0.57$, covered conductor $r_{\text{freq}} = 0.62$, undergrounding $r_{\text{freq}} = 1.00$. LOW (50% of MID) and HIGH (115% of MID) bands are computed as sensitivity bounds.

3.5.3 Cost Model

Annualized intervention cost is computed via Capital Recovery Factor applied to per-mile capital and operating expenditures:

$$\text{annual_cost} = \text{capex/mile} \times \text{miles} \times \text{CRF}(7\%, 30) + \text{opex/mile} \times \text{miles}$$

$$\text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1} = \frac{0.07 \times (1.07)^{30}}{(1.07)^{30} - 1} \approx 0.0806$$

Treated miles per cluster are tested at LOW = 1.0, MID = 2.0, and HIGH = 5.0 miles. The primary metric is:

$$\text{BCR} = \frac{\text{avoided_EAL}}{\text{annualized_cost}}$$

4. Risk Analysis

4.1 Portfolio Expected Annual Loss and Risk Concentration

The baseline portfolio EAL is **$277.9 million per year** (95% CI: **[$239.4M, $319.1M]**). This figure represents the expected cost of utility-linked wildfire ignitions across California under 2024 weather conditions. This is derived from public data without access to actual circuit maps, so this figure likely understates the true cost of pre-mitigation expected loss.

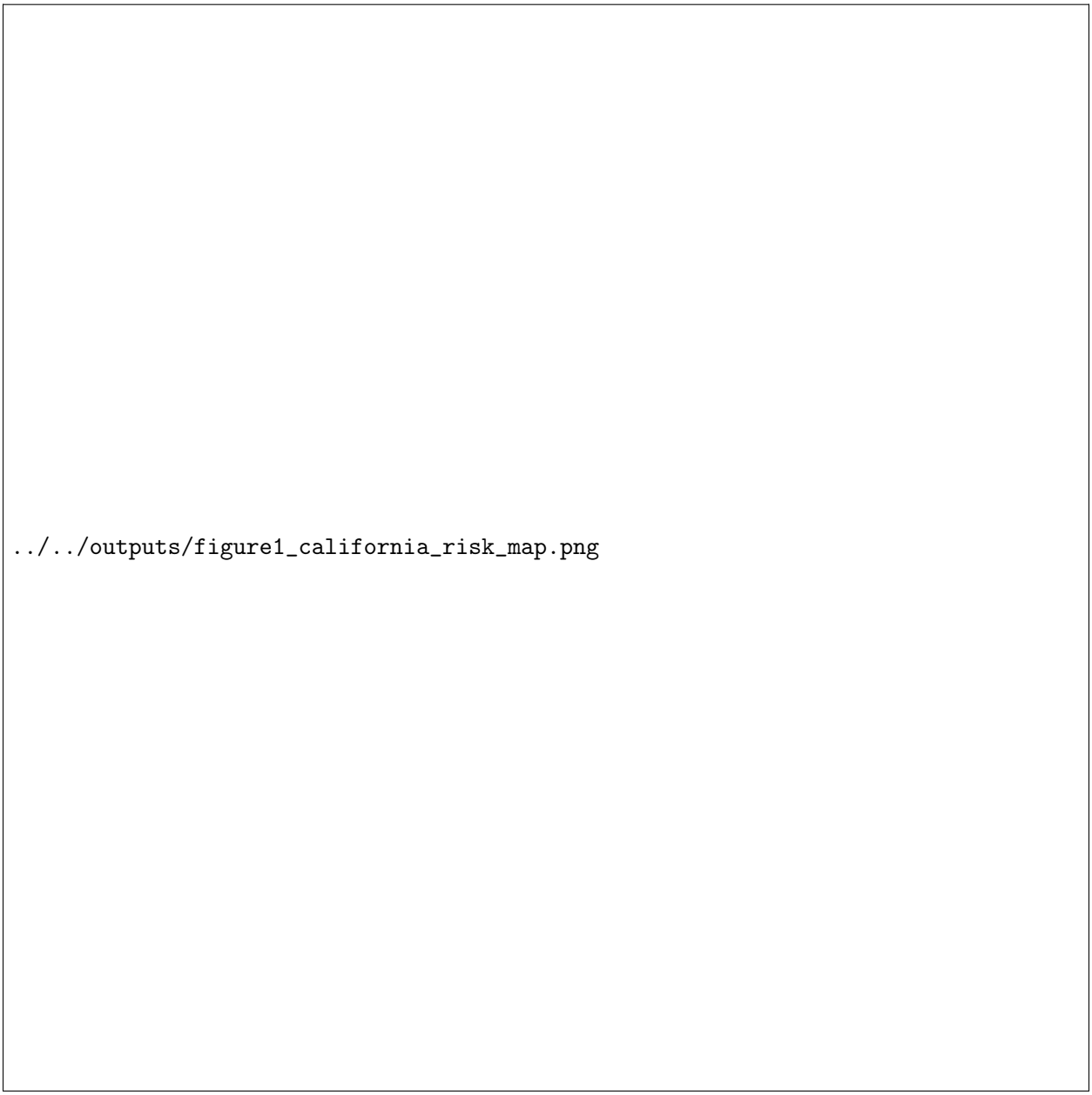


Figure 5: California Expected Annual Loss (EAL) by cluster. Circle size and color (log scale) encode base EAL. Black rings mark the top 28 highest-EAL clusters, which collectively carry 33.1% of portfolio EAL.

Expected annual loss is highly concentrated across the 1,000-cluster portfolio, following a distribution consistent with Pareto concentration:

Table 5: EAL concentration by cluster percentile.

Cluster Group	Cluster Count	EAL Share	Cumulative EAL
Top 5%	50	44.0%	$122.3M
Top 10%	100	59.3%	$164.9M
Top 20%	200	76.4%	$212.4M
Top 50%	500	96.0%	$266.8M
Bottom 50%	500	4.0%	$11.2M

The top 10% of clusters hold 59.3% of portfolio EAL with a concentration ratio of 5.93×. This extreme concentration is a key supporting finding of the study. Score-ranked allocation would cause the majority of mitigation capital to flow to clusters that carry only 4% of total expected loss. EL-ranked prioritization ensures capital flows first to the clusters carrying the greatest portion of total risk.

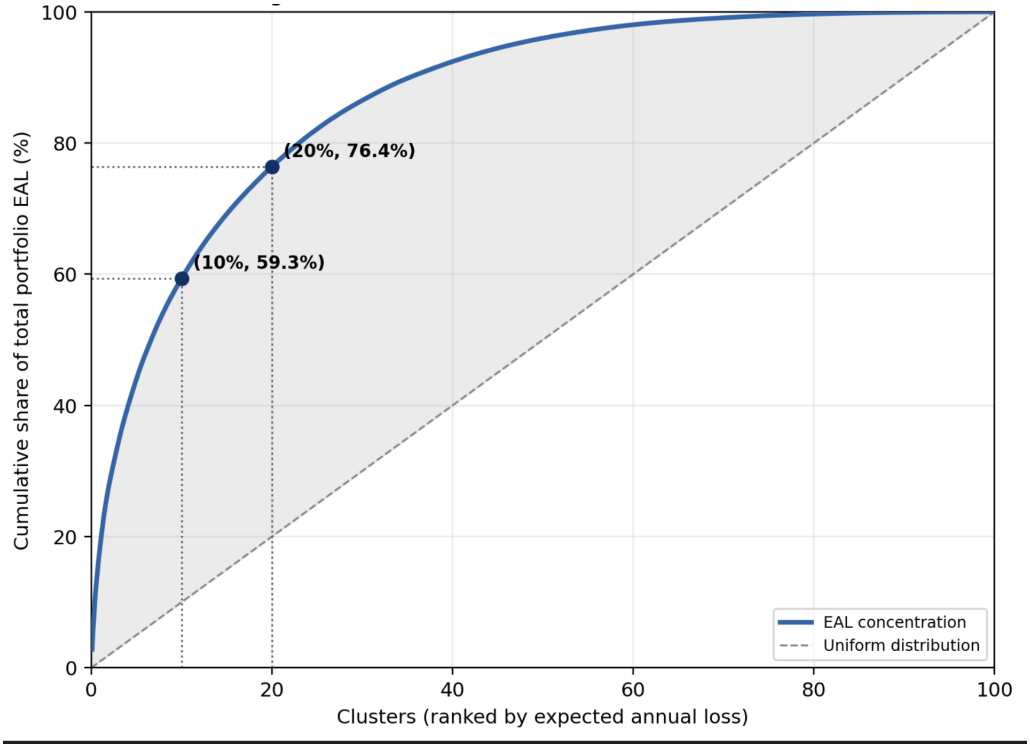


Figure 6: EAL concentration curve (Lorenz-type). The actual EAL curve shows high Pareto concentration: the top 10% of clusters carry 59.3% of total EAL. Gray shading represents the area between the EAL curve and uniform distribution.

4.2 Severity, Likelihood, Uncertainty, and Distribution

Likelihood. The ignition model achieves AUC = 0.795 on the held-out 2024 test year, with a 3.35× lift above the baseline prevalence rate. The top-10% cluster-day decile contains 34.2% of observed 2024 ignitions. Daily ignition probability for a given cluster ranges from near zero on cool, humid days to values

several hundred times the daily baseline on extreme fire-weather days (Burning Index $\geq$ 90th percentile, minimum relative humidity $\leq$ 10th percentile, wind speed elevated).

Severity. Conditional expected fire area given ignition ranges from 2.0 acres (nonburnable urban clusters subject to the plausibility gate) to hundreds of acres in Tier 3, shrub-fuel clusters under extreme weather. Large fires (exceeding 10 acres) drive the tail of the severity distribution: the P_{large} parameter ranges from 0.0 (nonburnable clusters) to approximately 0.35 in the highest-threat strata. The $500,000 per-structure damage assumption translates expected burned area to dollar loss.

Uncertainty. The 95% confidence interval for portfolio EAL is [$239.4M, $319.1M], showing variability in our cluster-level EAL estimates. Additional uncertainty sources such as severity prior specification, effect-size inputs, and how many miles are treated are addressed by the LOW/MID/HIGH sensitivity scenarios. Under LOW assumptions (1 mile treated per cluster), annualized cost is $12.0M and BCR is 5.52×. Under HIGH assumptions (5 miles), cost rises to $271.0M and BCR falls below 1.0 (0.49×), indicating that over-treating low-consequence clusters destroys value. The MID scenario ($50.5M annualized cost, BCR = 2.28×) is the median estimate.

Distribution. EAL is right-skewed across the cluster portfolio, with the top 5% of clusters (50 clusters) holding 44.0% of total EAL. This means that the expected-loss approach is most sensitive to correct identification and treatment of the highest-EAL clusters.

4.3 Impact on Parties

WUI Residents. A cluster-level ignition event a severity level in the large-fire category in a Tier 3 shrub-fuel zone with 500 structures/sq mi would destroy around 39–156 structures per event. Applying the average of 2.5 occupants per structure, this totals 98–390 individuals directly displaced per large-fire event. California has over 2.7 million structures in HFTD-designated zones [6], and the current annual expected structural loss overall is around 556 structures annually at the $277.9M EAL level. Reducing portfolio EAL by $115.3M (MID scenario) eliminates an estimated 231 structure-losses per year.

Ratepayers. The $3 billion annual IOU mitigation budget [12] finances $200 per ratepayer household per year in wildfire-specific mitigation charges. Under score-ranked prioritization, the EAL avoided per dollar spent is approximately 3.5× lower than under EL-ranked prioritization. This means that current prioritization methods require ratepayers to spend approximately $50.5M × 3.5 = $176.75M to achieve the same $115.3M in avoided EAL.

Utilities. The Wildfire Insurance Fund established under AB 1054 (2019) is capitalized at $21 billion jointly by the state and the three IOUs [25]. Our portfolio EAL of $277.9M/year implies that the fund would be depleted in 75 years under average conditions, but the Pareto concentration of risk means that a single catastrophic cluster event could generate losses of $500M–$2B, similar to the Camp Fire losses.

4.4 Scenario Analysis

Table 6: Cost-effectiveness scenarios under Warner et al. (2025) effect sizes.

Scenario	EAL Avoided	Ann. Cost	Net Benefit	BCR
LOW (1 mi/cluster)	$66.2M	$12.0M	+$54.2M	5.52×
MID (2 mi/cluster)	$115.3M	$50.5M	+$64.7M	2.28×
HIGH (5 mi/cluster)	$132.2M	$271.0M	−$138.8M	0.49×

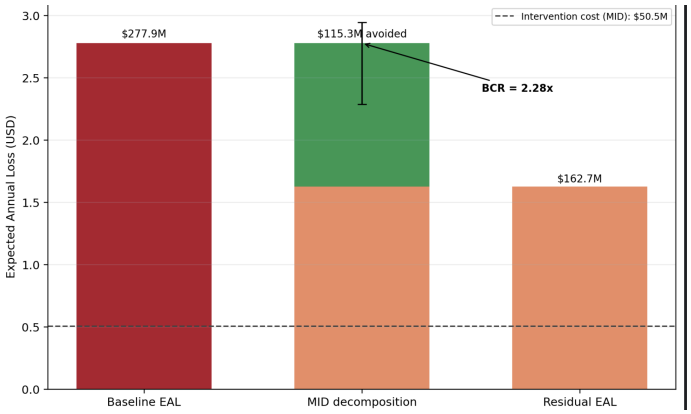


Figure 7: *

Figure 7: (a) Baseline vs. MID intervention counterfactual

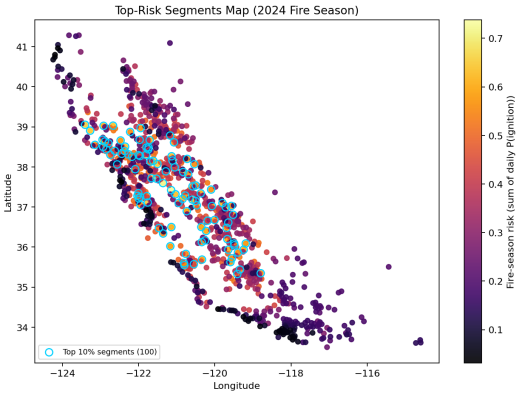


Figure 8: *

Figure 8: (b) Top-100 cluster EAL map (2024)

Figure 9: Counterfactual analysis. (a) Baseline EAL ($277.9M) vs. MID scenario: $115.3M avoided, $50.5M annualized cost, BCR = 2.28×. (b) Top-100 highest-EAL clusters (cyan rings) targeted under EL-ranked framework.

4.5 EL-Ranked vs. Score-Ranked Prioritization

Table 7: EL-ranked versus score-ranked allocation under equal annual budget (MID scenario).

Metric	EL-Ranked	Score-Ranked
Annual budget	$2.84M	$2.84M
Clusters selected	25	45
EAL avoided	$58.7M	$16.6M
BCR	20.71×	5.91×

4.6 Known Limitations

Four limitations are material to interpreting these results. First, we must address us using clusters as a proxy for granular electrical data. Clusters are constructed from incident locations, so areas where existing mitigation has already eliminated ignitions contribute no clusters and no labels. This implies the model is biased towards areas where ignitions have already occurred. Additionally, the entire electrical grid is not available to the public, so we did not have access to the wealth of data which could have been used to train a significantly stronger model as was done in similar studies. Next, the severity of loss layer is a physical fire-spread model, so our estimates are not as specific as they could be. In the interest of computational time and the complexity of this project, we simplified by using an average value of $500,000 dollars. Third, our intervention effects are scenario inputs imported from Warner et al. (2025) [14] and are not causally estimated from this dataset.

5. Recommendations

5.1 Primary Recommendation: Adopt EL-Ranked Intervention Allocation

California's current utilities should consider adopting an approach where the loss prioritization multiplies ignition probability with the severity of the fire rather than their current composite score prioritization. We address this recommendation specifically to: CPUC, the Office of Energy Infrastructure Safety, the management teams of PG&E, SCE, and SDG&E. The composite score approaches, instead of multiplying the two components, add them together. This better represents the fact that expected loss is inherently multiplicative. For instance, a scenario in which a certain segment had high-ignition-frequency but low consequence would have the same expected loss as a segment with low-ignition-frequency but high consequence even though the latter would result in significantly more damages. Failure to use the expected value in densely populated areas can be especially damaging and dramatically increase the size of the losses.

When projects are ranked by expected loss under the MID scenario, about $50.5M million per year is directed to the most critical clusters, reducing nearly $115.3M in expected annual loss (Benefit to Cost-Ratio = 2.28). On the other hand, if the same budget was distributed under the current score-based rankings, the reduction totals only 16.6M. In other words, relying on composite scores results in a nearly 3.5 time drop in efficiency. From the perspective of ratepayers, this translates to roughly $126M in additional costs each year to achieve similar reduction in wildfire risk.

5.2 Intervention Priority by Cluster Type

Fast-Trip Settings and Hardening (222 clusters): For clusters with EAL exceeding $1.0M/year in HFTD-designated zones, Enhanced Powerline Safety Settings (EPSS) in addition to equipment hardening make up the recommended baseline action. Warner et al. (2025) documents a 72% ignition frequency reduction at the lowest cost per ignition avoided of any evaluated intervention type [14]. This type of intervention is most impactful for WUI residents in these clusters, who face the highest probability of fire exposure from utility equipment on an average day.

Enhanced Vegetation Management (106 clusters): For mid-tier clusters (EAL between $200K and $1.0M/year) in burnable LANDFIRE fuel categories, improved vegetation management gets a 57% frequency reduction. This application is, however, restricted to clusters where the fuel category confirms that the vegetation could cause a fire. EVM directly reduces the severity of fires that do occur by removing continuous fuel sources that allow for rapid spread of wildfires due to the presence of constant fuel.

Covered Conductor (4 clusters): For clusters where residual EAL remains increased after applying fast-trip assumptions, covered conductor at $480K/mile provides additional protection. Only 4 clusters meet this threshold under conservative assumptions.

Targeted Monitoring (668 clusters): For specific clusters where the expected yearly loss is lower than the annual cost threshold, we do not suggest a specific intervention. It is best to keep optionality open through constant monitoring methods including infrared sensing, enhanced patrol schedules, and weather-contingent inspections.

Undergrounding (0 clusters): Under conservative EAL assumptions, no cluster clears the $2.6M/year threshold that we believe justifies undergrounding at $3.0M/mile. The top 28 clusters by EAL (33.1% of portfolio risk) are recommended on a per basis feasibility review if cost or consequence assumptions are revised. We do not overtly discourage undergrounding, but instead recommend considering alternative approaches first since the public data available do not provide a sufficient enough basis to justify undergrounding for any singular region.

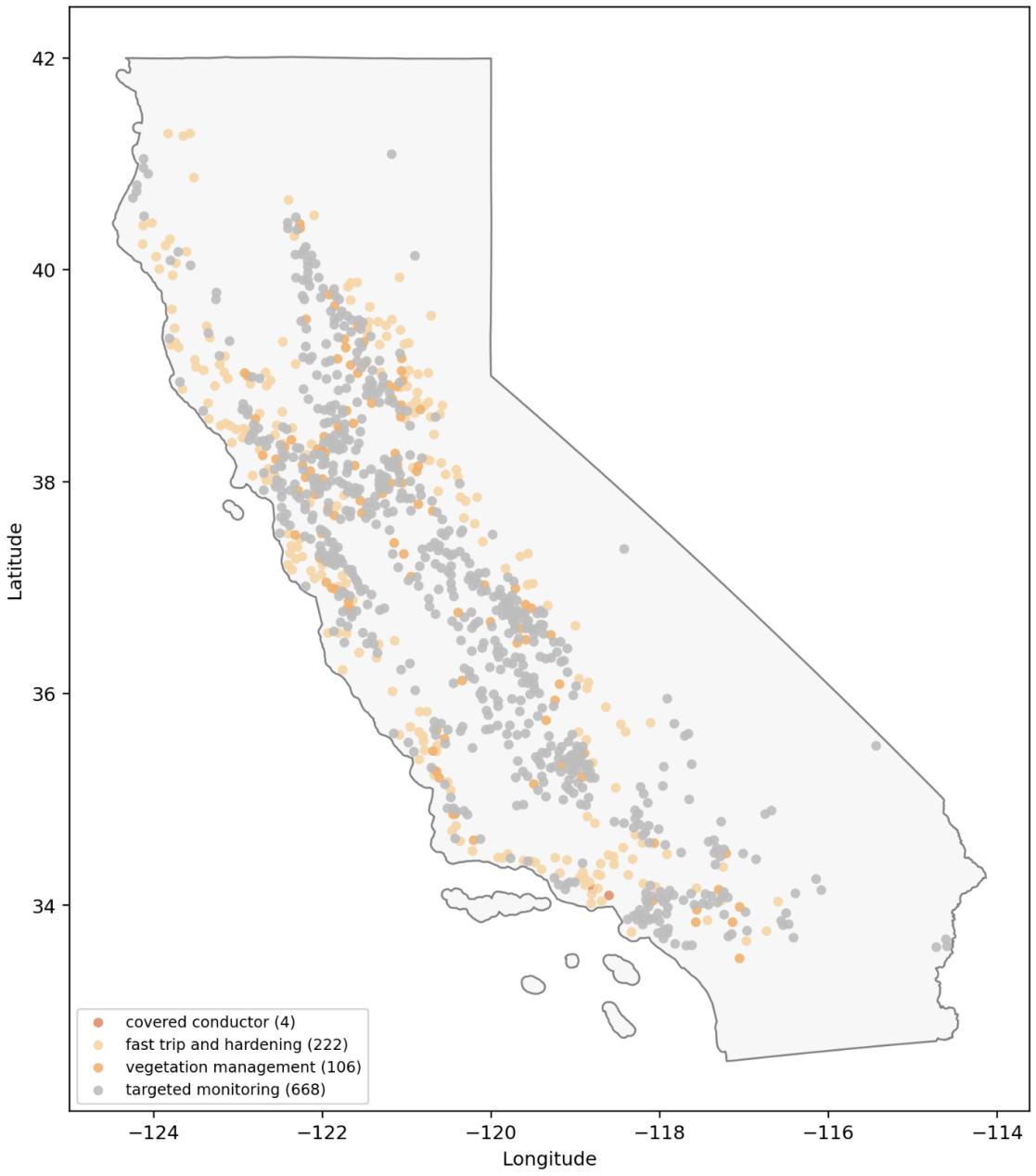


Figure 10: Spatial allocation of recommended intervention classes across clusters under the MID scenario.

5.3 Risk Mitigation Strategy Analysis

Insurance. California Assembly Bill 1054 (2019) established the $21 billion Wildfire Insurance Fund which was jointly funded by the state and the three IOUs in order to cover utility liability that would exceed individual balance sheet capacity [25]. This fund works as insurance: utilites would pay yearly contributions based on their wildfire risk exposure. The actuarial capacity of this fund would depend directly on the portfolio EAL throughout the utility service areas. If the contributions are instead based on the the score-ranked risk assessments which underestimate the expected loss especially in high consequence

WUI segments, the fund would be undercapitalized relave to its true liability exposure. Our model's $277.9M yearly EAL estimate instead provides an independent benchmark from which the funds used can be evaluated. In addition, the concentration finding (top 10% of clusters carrying 59.3% of EAL) provides information towards reinsurance structuring: reserves should be comparable in size to the tail of the cluster EAL spread rather than the mean. Ensuring the accuracy of the capitalization methodology would directly benefit ratepayers (who fund the IOU contributions) and WUI residents (who depend on the fund's solvency to receive claims.

Behavior Change. Utilities can apply our ignition model towards Enhanced Powerline Safety Settings (EPSS). EPSS is a protocol where utility control room staff determine what circuits to put into fast trip mode based on Red Flag Warnings in the area they are considering. The cluster-day ignition probability scores are necessary to utilize EPSS in the sense that instead of applying fast trip to entire transmission lines, utility staff can target specific high-EAL clusters in which there is a higher probability of ignition on a specific day.

Modifying Outcomes. The enhanced vegetation management (EVM) strategy reduces the extent of spread from an ignition point by removing fuel types that may cause rapid spread, causing greater loss. Warner et al. (2025) estimate a 57% frequency-reduction effect for EVM, showing its value in managing the 106 shrub and timber fuel clusters where both ignition probability and possible spread potential are relatively high [14]

5.4 Operational Implementation

This framework is designed to be used multiple times, getting re-calibrated each time, rather than one-time deployment. The recommended operational workflow:

1. Refresh CPUC incident reports and gridMET weather data quarterly.
2. Re-score all 1,000 clusters across the upcoming 90-day planning horizon.
3. Re-rank clusters by updated EAL estimate.
4. Allocate available intervention budget to top-ranked untreated clusters.
5. Track avoided ignitions quarterly; recalibrate effect-size assumptions against observed outcomes.

6. Conclusion

The risk of wildfire from utility infrastructure is a risk that can be easily mitigated with a specific resource planning and allocation process. The key finding of this research we would like to highlight is that 115.3 million in expected annual loss at an annual cost of $50.5 million (BCR = 2.28×, MID scenario) would be saved through an expected-loss framework.

WUI residents: The proposed program is expected to avert 231 losses annually under the MID scenario, with a focus on key regions where people are at the highest risk of injury. Families in Tier 3 fuel zones, which include areas around the Sierra Nevada foothills, coastal ranges, and Southern

California mountain areas, will be the main beneficiaries of the rapid deployment program and vegetation management in 328 clusters that have been identified for active intervention.

Ratepayers: shifting from score-based ranking to an expected-loss framework delivers the same $115.3 million reduction in annual expected losses with $50.5 million in annualized spending. Achieving that same level of risk reduction under the current score-ranked approach would require roughly $176.75 million. The difference—about $126 million per year—represents avoidable mitigation costs brought on by electricity customers purely because of the prioritization method. This gap highlights the need for regulators to reassess how wildfire mitigation investments are ranked and allocated.

Utilities: correctly identifying and alleviating the top-EAL clusters dramatically reduces the probability of the ignition events that have historically been economically devastating. The Wildlife Insurance Fund's long-term actuarial survival depends on accurate EAL estimation. Correctly identifying areas where more funds are needed and areas where less funds are needed result in reduced chances of liabilities for utilities.

A more cost-effective standard: this study demonstrates that using an expected-loss-based WMP prioritization rather than the score-based one traditionally used has multiple benefits. In particular, this approach is both techincally viable and economically beneficial. One constraint on this approach is the limitations of data access. Having greater and reliable access to geospatial data and standardized BCR reporting could close the gap and increase the reliability and feasibility of such an approach.

Wildfire risk in California will continue to climb as climate conditions become more fragile due to global warming. The tools to allocate mitigation resources more effectively already exist, but are not utilized just yet. Adopting an expected-loss framework in WMP prioritization is not a marginal improvement, but is the methodologically correct approach, and through the implementation of a cheaper, smarter alternative in contrast to throwing money at the problem and hurting ratepayers, we can mitigate the threat of wildfires on the west coast.

Appendix A: Extended Model Diagnostics

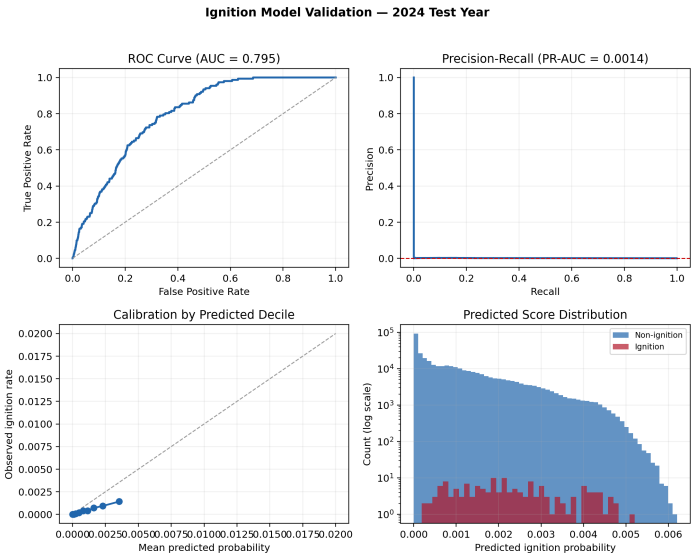


Figure 11: *

(a) Ignition model validation (2024 held-out test year)

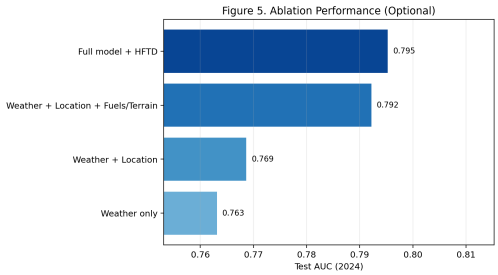


Figure 12: *

(b) Incremental AUC by feature layer

Figure 13: Ignition model diagnostics. (a) ROC curve (AUC = 0.795), precision-recall curve, calibration by predicted decile, and score distribution by outcome class. (b) Marginal AUC contribution of each feature layer on held-out 2024 data.

A.1 Full Ablation Study Results

Table 8: Full ablation results on held-out 2024 test data.

Experiment	Feature Layers Included	AUC (2024 Test)
1. Weather only	BI, ERC, rmin, wind, tmax, precip, month, year, season	0.763
2. + Location	Cluster centroid (lat, lon)	0.769
3. + Fuels & Terrain	Elevation, slope, fuel category, fuel code	0.792
4. + HFTD (full model)	HFTD regulatory tier	0.795

A.2 Severity Specification

In the 1,000-cluster run, all 19 lookup cells are populated from utility-anchored levels: 6 direct (HFTD $\times$ fuel), 3 fuel-only, and 10 HFTD-only. FRAP fallback was not required in this run.

Appendix B: Scenario and Budget Sensitivity

Table 9: LOW/MID/HIGH cost sensitivity under Warner et al. (2025) effect sizes.

Scenario	EAL Avoided	Ann. Cost	Net Benefit	BCR
LOW (1 mi/cluster)	\$66.2M	\$12.0M	+\$54.2M	5.52×
MID (2 mi/cluster)	\$115.3M	\$50.5M	+\$64.7M	2.28×
HIGH (5 mi/cluster)	\$132.2M	\$271.0M	−\$138.8M	0.49×

Table 10: EL-ranked versus score-ranked under equal annual budget (MID scenario).

Metric	EL-Ranked	Score-Ranked
Annual budget	\$2.84M	\$2.84M
Clusters selected	25	45
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BCR	20.71×	5.91×

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